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|RESEARCH ARTICLE

Knowledge of women of childbearing age according to their exposure to awareness-raising activities on feeding practices during the critical period of the first 1000 days in urban Niamey

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|Abstract

Background: Undernutrition and micronutrient deficiencies among women of childbearing age (WCA) remain a major problem in Niger. The MERIEM project, implemented by GRET in Niamey, has supported the local production and marketing of fortified products, in particular Foura Soga for WCA, while raising awareness among WCA of good food and nutrition practices.

Aims: The aim of this study is to assess the evolution of WCAs' knowledge and practices in terms of nutrition as a function of their exposure to awareness-raising activities on good nutrition practices.

Patients and Methods: A cross-sectional survey was carried out among 856 WCA in 780 households in the 5 municipalities of Niamey, using a standard five-module questionnaire administered at home. The data, processed with R and STATA, were used to construct knowledge and awareness scores based on the mother's answers to the questions asked. Exposure scores were also produced according to women's level of participation in nutrition education activities. Results are presented according to physiological and consumption status, and adjusted for household wealth quintiles.

Results: The results of this study showed a low score (1/8) for participation in nutrition awareness and education activities. Female consumers had significantly higher knowledge scores ($p<0.001$). The study noted a positively significant association between participation in activities and the level of household exposure ($p<0.001$). On the other hand, women's level of exposure to activities was low and significantly dependent on household standard of living ($p<0.05$).

Conclusions: This study highlights the positive impact of awareness-raising and nutrition education activities on WCAs' knowledge of good nutrition practices.

Keywords: Nutritional knowledge|Dietary practices |1000-day period | Nutrition education | Women of childbearing age| Niamey

1|Introduction

Undernutrition and nutritional deficiencies remain major public health problems worldwide, particularly among the most vulnerable populations (UN Nutrition, 2021). According to the World Nutrition Report, nearly 149 million children under the age of five were stunted in 2023, as a direct consequence of inadequate nutritional intake, especially during critical periods of development (FAO, 2023). Sub-Saharan Africa remains one of the most affected regions, with a persistent prevalence of infant and maternal malnutrition (Van & Cornelia, 2018; Teshale et al., 2025). Niger in particular has worrying nutritional indicators, despite a slight improvement in recent years (INS, 2022). This is due to a number of factors, including poverty, food insecurity, low levels of education among mothers and inappropriate dietary practices (INS, 2024).

Faced with this situation, the State of Niger, in collaboration with its technical and financial partners, has implemented various programs aimed at improving the nutritional situation of the population and combating undernutrition and its long-term harmful consequences. These initiatives include multi-sectoral actions such as the National Nutritional Security Policy (PNSN) and the implementation of community projects focused on preventing malnutrition. These efforts have strengthened health services, promoted exclusive breastfeeding and improved nutritional knowledge through awareness campaigns (HC I3N, 2021).

The first 1000 days of life “from conception to the child's second birthday” are now recognized as a critical window for the prevention of malnutrition and optimal child development (Ramirez-Rozzi et al., 2021; Akolly et al., 2021). Adequate nutrition during this period influences not only physical and cognitive growth, but also the long-term health of both child and mother (Morcel et al., 2023). Numerous studies have shown that nutritional interventions targeting this phase have lasting effects on countries' human capital and socio-economic development (Iknane et al., 2011; Fadare et al., 2019; Appoh et Krekling, 2005). Maternal knowledge of nutrition and health practices plays an essential role in the evolution of children's health, general development and cognitive growth (Teshale et al., 2025; Prasetyo et al., 2023; Forh et al., 2022). Nutritional education for mothers can alleviate malnutrition and deficiencies in their young children (Jardí et al., 2021).

This is the perspective to the MERIEM Project (Mobilizing Sahelian Enterprises for Innovative Large-Scale Responses to Malnutrition), which aims to improve nutritional practices through community education and awareness-raising among women of childbearing age.

This study focuses on the impact of the MERIEM project's awareness-raising and nutrition education activities on women of childbearing age's knowledge of feeding practices during the first 1,000 days.

Numerous studies have explored mothers' knowledge, attitudes and practices on infant nutrition in Niger, but they often neglect the complex nuances of maternal exposures to nutrition awareness activities. For this reason, in this study we aim to understand the extent to which nutrition education activities contribute to improved nutritional behaviours, a key factor in the sustainable reduction of maternal and child malnutrition in Niger.

The aim of this study is to assess changes in the nutritional knowledge and practices of women of childbearing age as a function of their exposure to awareness-raising activities.

2| Material and Methods

2.1|Geographical location

The study took place in the 5 municipalities of the city of Niamey, in 39 randomly selected counting area covering some 30 neighborhoods.

2.2|Target population and stratification

The study's target groups were women of childbearing age (WCA) aged 15 to 49 in the city of Niamey. The sample was stratified according to women's physiological and consumption status. We therefore included pregnant women (including pregnant and breast-feeding women), breast-feeding women (excluding pregnant and

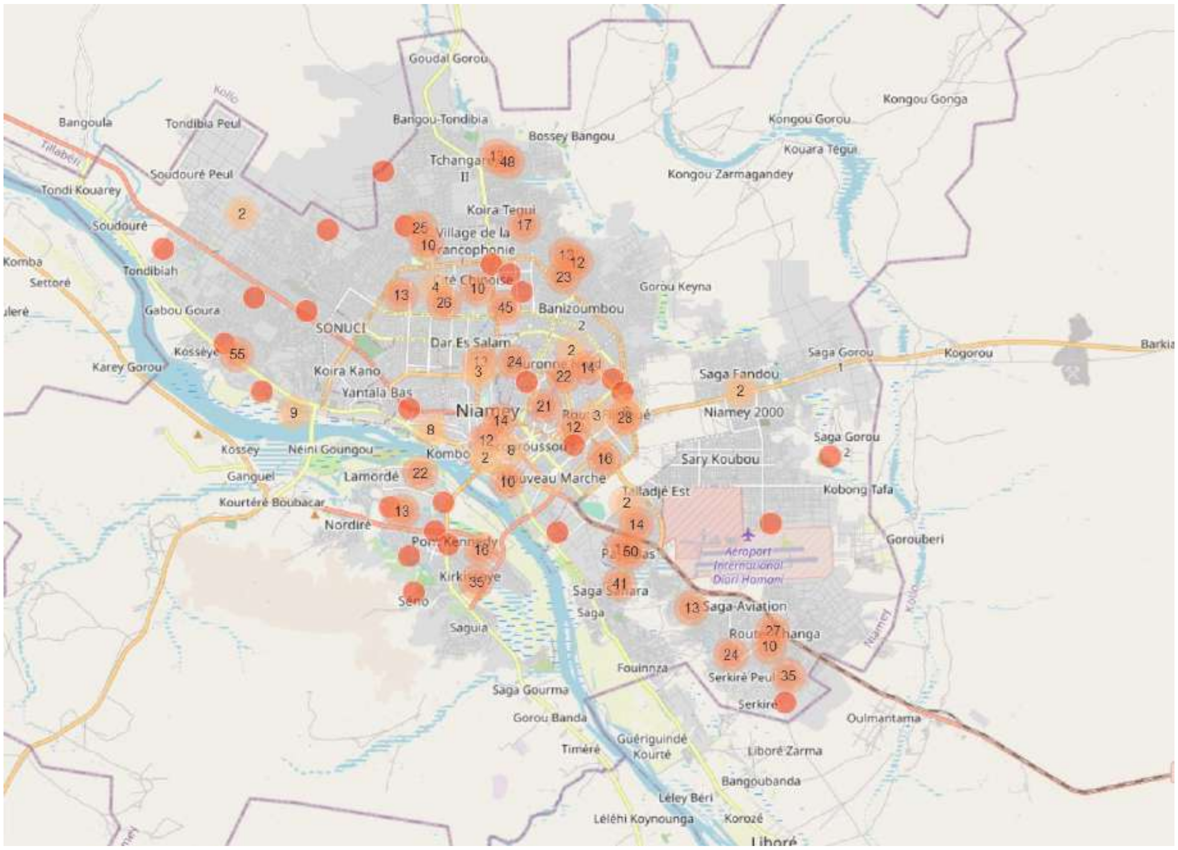

Figure1: geographical location of the study site

breast-feeding women) and women who were neither pregnant nor breast-feeding. Women Consumers are those who consumed the fortified product foura soga at least once during the previous week, the 7 days prior to the visit of the interviewers. Non-consumers were those who did not meet the “consumer” criterion, those who had not consumed Foura Soga in the previous week.

2.3 | Sample size

Sample size was calculated according to the following formula (Hayes & Bennett, 1999):

$$n=(1+\lambda)\times Deff\times(2\sigma^2 \quad (1-\rho) \quad [Z_{(1-\alpha/2)}+Z_{(1-\beta)}]^2)/ \left[(X_2-X_1)\right]^2$$

Where:

n= Sample size

Deff= Effect of sample design

λ= Percentage of data loss

σ^2= Variance

ρ= Correlation coefficient

X_1= Estimated level of an indicator at the time of the first survey (baseline)

X_2= Expected level of the indicator, at a future date, for the project area so that the quantity (X_2-X_1) is the size of the magnitude of change we wish to be able to detect

$Z_{(1-\alpha/2)}$= the Z-score corresponding to the degree of confidence, for the index (1-α/2), that we wish to have to conclude that an observed change in size (X_2-X_1) is not due to chance (statistical significance)

Z_(1-β)= the Z score corresponding to the degree of confidence, for the index (1-β), that we wish to have to detect with certainty a change in size (X_2-X_1) if such a change has indeed taken place (statistical power).

The objective was to detect, when it exists, a difference ε of a specified indicator, between the comparison groups. The sample size for the survey was 856 women, distributed proportionally across the different municipalities of the city of Niamey, taking into account their physiological status.

2.4 | Collection tool

Data were collected using a standardized questionnaire with several modules. It included both closed questions (true/false? at what age...?) and open questions focusing on a number of good nutritional practices (FAO, 2016). Through this questionnaire, women were tested on their knowledge concerning: (1) Breastfeeding (module R); (2) Complementary feeding of infants and young children (module S); (3) Nutrition for pregnant and breastfeeding women (module T); (4) Vitamins, minerals, deficiencies (module U) and fortified products (module V).

Data were collected in electronic format from August 10 to 16, 2022, using touch-screen tablets and the ODK digital application.

2.5 | Wealth quintiles

The socioeconomic status index for households in Niger is usually constructed using data from the Living Standards Measurement Study (LSMS). However, in the absence of complete data, information collected on the quality of housing and household assets was used to construct a proxy for this indicator. A total of 25 variables were used, including 10 variables on housing quality and 15 variables on assets owned. The variables selected for constructing the score are categorical, and the Multiple Correspondence Analysis (MCA) method is ideal for summarizing the information they contain. Two factor axes containing sufficient information were selected and used to construct the Household Living Standards Proxy Index (PIBEE) (Filmer & Pritchett, 2001; Van & Cornelia, 2018). Households were then ranked in order of increasing wealth score (from least affluent to most affluent) and divided into quintiles (each quintile corresponds to 20% of households) ranging from Q1 (least affluent) to Q5 (most affluent) (Rutstein, 2008). Wealth quintiles were created to categorize households according to their GDPEE values (Hewett & Montgomery, 2001).

2.6 | Creation of knowledge scores

Based on the answers given to the various questions in the knowledge section of the questionnaire, a points system was devised, and each woman was then given a score. Eight scores, varying according to question and response mode, were thus constructed as follows:

- ⇨ 9 points for knowledge of breastfeeding practices at birth corresponding to 10 questions (M1 to M10);
- ⇨ 8 points for knowledge of complementary feeding of infants and young children (N1 to N11, questions N6, N8 and N9 are excluded from the scores);
- ⇨ 7 points for knowledge of the nutritional status of young children (YC) corresponding to 4 questions from N12 to N15;
- ⇨ 4 points for knowledge of the diet of pregnant women corresponding to 5 questions (O02 to O05, question O01 excluded);

- 4 points for knowledge of the diet of breastfeeding women corresponding to 4 questions (O06 to O09);
- 8 points for knowledge of vitamins and minerals and nutritional deficiencies corresponding to 5 questions (P3 to P7);
- 16 points for knowledge of fortified products corresponding to 9 questions from P9 to P17;
- 15 points for participation in awareness-raising activities on nutrition issues corresponding to 6 questions from P18 to P23.

2.7|Creating exposure scores

On the basis of the number of participations awareness-raising activities and the type of activity in which the women had taken part, we assigned exposure scores to each woman. The maximum score for awareness-raising activities was 5.7 points.

These scores enabled us to differentiate women according to their level of exposure to the awareness-raising activities. Thus, women are divided according to their scores into the following 4 groups (Table1).

2.8|Data analysis

The data were analyzed using the appropriate statistical software packages “R” and “STATA”. The statistics were mainly descriptive (frequency, mean, median, standard deviation, standard error and proportions) and enable us to assess the distribution (extreme values, outliers or missing values, quintiles) and characteristics of the various variables in the study. Various analyses (multivariate descriptive analyses) and statistical tests (contingency table) were also carried-out to search for possible associations between variables.

2.9|Ethical considerations

Participation in the study was free and voluntary. Local administrative and customary authorities were informed of the objectives and conduct of the study. The study was conducted in such a way that “taboos”, traditions and “customs” were considered and respected.

Individuals selected to take part in the survey were informed of the voluntary nature of their participation. It was made clear to them that they were free to refuse to participate. If they agreed to take part in the study, however, they could withdraw at any time, postpone the investigators' visits or refuse to answer certain questions, without having to justify their decision.

Written consent was signed by the respondent and the head of household.

3| Results

3.1| Knowledge scores of women of childbearing age on good food and nutrition practices during the first 1,000 days of life

In the course of this study, 856 women of childbearing age between 15 and 49 were surveyed in the 5 municipalities of the city of Niamey. Table 1 shows that breastfeeding women were more represented in the sample (52%), compared with 36% for women who were neither pregnant nor breastfeeding, while pregnant women represented only 9% of the total sample. The distribution of average knowledge scores on good breastfeeding practices among surveyed women by physiological status shows a relatively average level of knowledge across all physiological classes considered. The average total score was 4.41 ±1.75 out of a maximum of 9 points, and varied very little according to physiological status. Only breastfeeding women had a higher mean score than the overall average (4.54 ± 1.67), although this difference was not statistically

Table 1. Classification of exposure scores for awareness-raising activities

Awareness level MERIEM	Scores
Not sensitized	0
Low sensitivity	[0.2-1.9]
Moderately sensitized	[2-3.9]
Highly sensitized	≥ 4

significant. In terms of mean scores for knowledge of complementary foods and good complementary feeding practices, the mean score was 2.72 ±1.29 out of a maximum of 8 points. The women's level of knowledge about complementary foods was relatively low, with no woman in any category scoring above the average (4/8). On the other hand, the women surveyed have a good knowledge of their children's nutritional status, with regard to the signs and forms of malnutrition. The average score for knowledge of nutritional status and child nutrition during and after illness was 5.63 ±1.75 out of a maximum of 7 points. Pregnant women's average knowledge of their own diet was 2.04 ± 0.75 out of 4 points. Breastfeeding women, on the other hand, have below-average knowledge of their own diet 1.75 ± 0.77 out of 4 points. Knowledge of vitamins, minerals and nutritional deficiencies is lower among the women of childbearing age surveyed, both by name and by potential food sources, with an average score of 1.41 ± 1.79 out of 8 points. Women also participate less in awareness-raising activities on nutrition issues, whether as part of the MERIEM project or other programs. The average participation score is less than 2/16. Breastfeeding women seem to participate more in these activities than pregnant women and women who are neither pregnant nor breastfeeding. Fortified products are also poorly known, according to the average knowledge score of 2.54 ±3.32 out of a total score of 15 points (Table2).

Table 2. Repair of women's knowledge scores according to their physiological state

Knowledge score	P.W (97)	B.W (448)	NPNBW (311)	P-value[2]	Total (856)[1]
Breastfeeding	4.27 (1.80)	4.54 (1.67)	4.28 (1.83)	0.2	4.41 (1.75)
Supplementary feed	2.72 (1.44)	2.74 (1.23)	2.69 (1.33)	0.9	2.72 (1.29)
Nutritional status	5.41 (1.91)	5.59 (1.75)	5.76 (1.69)	0.3	5.63 (1.75)
Pregnant woman (P.W)	2.04 (0.75)	-	-	-	2.04 (0.75)
breast-feeding woman (B.W)	0.01 (0.10)	1.75 (0.77)	-	<0.001	0.92 (1.04)
Vitamins and minerals	1.45 (1.68)	1.44 (1.83)	1.35 (1.78)	0.6	1.41 (1.79)
Nutritional awareness	1.68 (1.85)	2.12 (1.90)	1.88 (1.93)	0.012	1.98 (1.91)
Fortified products	2.37 (3.71)	2.56 (3.26)	2.56 (3.30)	0.4	2.54 (3.32)

[1] Average (SD)

[2] Kruskal-Wallis test

The breakdown according to consumption status of the fortified product foura soga gives 10% female consumers and almost 90% female non-consumers. The difference is statistically significant (P=0.005) according to physiological status, using the chi[2] test. Generally speaking, women who use the fortified product Foura Soga have better knowledge scores than their non-user counterparts. Differences in knowledge were statistically significant for knowledge of vitamins and fortified products, and for participation in nutrition awareness activities. Respectively P=0.011; P<0.001; P<0.001 according to the Wilcoxon-Mann-Whitney test.

Table 3 summarizes the distribution of knowledge scores according to household wealth quintiles. An increasing consumption of the Foura Soga product by the Q1 quintile (- affluent) towards the Q5 quintile (+ affluent) is to be noted. The percentages decrease progressively from 31% for quintile Q5 to 6.4% for quintile Q1, and the difference is highly significant according to the non-parametric chi2 test (p=0.003). The affordability of the Foura Soga is therefore clearly linked to the socio-economic level of the households in which the women live. According to the Kruskal Wallis test, most of the knowledge scores established (except the score on breastfeeding women's diet) increase significantly with wealth quintile from the least affluent households (Q1) to the most affluent (Q5). The higher the household's socio-economic standard of living, the better the knowledge scores on good infant and maternal nutrition practices (Table 3). Participation scores in nutrition awareness activities are also positively correlated with household wealth quintiles. Thus, the higher the household's socio-economic standard of living, the more women in the household participated in activities to raise awareness and promote good nutrition (p<0.05, Kruskal-Wallis test).

Analysis of the table of WCA knowledge scores according to level of exposure to MERIEM project awareness-raising activities (Table 4), shows that high exposure to awareness-raising considerably improves nutrition knowledge scores. In other words, the more women are exposed to MERIEM awareness-raising activities, the better their knowledge scores on nutrition and food issues. Knowledge scores were significantly better among highly sensitized women (P<0.001).

Table 3: Distribution of women's knowledge scores by wealth quintile

Knowledge score	Q1[1](156) *(Poor)*	Q2[1](156)	Q3[1](156)	Q4[1](156)	Q5[1](156) *(Riche)*	p-value[2]
Breastfeeding	4.15 (1.77)	4.10 (1.66)	4.36 (1.63)	4.68 (1,60)	4.90 (1.79)	<0.001
Supplementary_feed	2.34 (1.26)	2.67 (1.28)	2.76 (1.31)	2.85 (1,22)	2.96 (1.34)	<0.001
Nutritional status	5.23 (1.75)	5.38 (1.85)	5.53 (1.64)	6.15 (1,58)	5.75 (1.75)	<0.001
Pregnant_woman (P.W)	0.30 (0.72)	0.17 (0.58)	0.17 (0.60)	0.23 (0,73)	0.35 (0.85)	0.051
breast-feeding woman (B.W)	0.95 (0.98)	0.93 (1.01)	0.93 (1.05)	0.98 (1,10)	0.82 (1.09)	0.5
Vitamins and minerals	1.38 (1.68)	2.15 (3.01)	2.32 (3.10)	3.29 (3,85)	3.55 (4.06)	<0.001
Nutritional awareness	1.06 (1.66)	1.12 (1.69)	1.43 (1.82)	1.71 (1,88)	1.75 (1.79)	<0.001
Fortified products	1.52 (1.55)	1.85 (1.71)	1.97 (1.75)	2.28 (2,07)	2.48 (2.28)	0.001

[1] Average (SD)
[2] Kruskal-Wallis test

3.2 | Exposure scores for awareness-raising activities on good feeding and nutrition practices during the first 1,000 days of life

There was no statistically significant relationship between women's physiological status and participation in the MERIEM project's awareness-raising activities. All women had approximately the same level of exposure to all nutrition education activities. However, breastfeeding women seem to participate more in these activities than pregnant women and women who are neither pregnant nor breastfeeding (Table5).

Table 4: Women's knowledge scores according to their exposure status to MERIEM awareness-raising activities

Knowledge score	Not sensitized (304[1])	Low sensitivity (467[1])	Moderately aware (78[1])	Highly sensitized (7[1])	P-value[2]
Breastfeeding	4.04 (1.82)	4.49 (1.68)	5.29 (1.40)	5.57 (1.81)	<0.001
Supplementary feed	2.48 (1.29)	2.86 (1.23)	2.81 (1.50)	2.43 (1.45)	<0.001
Nutritional status	5.48 (1.80)	5.52 (1.72)	6.78 (1.30)	7.00 (1.00)	<0.001
Pregnant woman (P.W)	0.26 (0.72)	0.21 (0.66)	0.21 (0.74)	0.71 (1.25)	0.2
breast-feeding woman (B.W)	0.86 (1.00)	0.90 (1.02)	1.31 (1.23)	0.86 (1.07)	0.033
Vitamins and minerals	1.05 (1.73)	1.47 (1.78)	2.42 (1.70)	2.00 (1.91)	<0.001
Nutritional awareness	0.43 (0.50)	2.25 (1.22)	5.90 (1.35)	7.86 (0.38)	<0.001
Fortified products	1.90 (2.52)	2.55 (3.23)	4.66 (4.95)	5.86 (6.28)	<0.001

[1] Average (SD)

[2] Kruskal-Wallis test

The level of exposure of the women of childbearing age surveyed to the MERIEM project's awareness-raising activities on child and women's nutrition was low. Consumers of foura soga were much more exposed to the activities than non-consumers, which could explain their reasons for consuming the product. The difference is strictly significant according to the non-parametric Wilcoxon test, whatever the activity considered (Table 5). Women were much less exposed to MERIEM awareness-raising activities on nutrition, such as film screenings followed by debates and awareness-raising sessions in integrated health centers 0.13 ± 0.35 over 1.7 points max.

Table 5. Awareness-raising exposure scores according to women's physiological status

Awareness-raising activities	PW (97)	BW (448)	NPNBW (311)	P-value[2]	Total (856)[1]
Participation in MERIEM nutrition awareness activities	0.10 (0.33)	0.14 (0.37)	0.12 (0.33)	0.2	0.13 (0.35)
Fati MARIKO's song on nutrition for pregnant women and young children	0.42 (0.50)	0.43 (0.50)	0.41 (0.49)	>0.9	0.42 (0.49)
Promotional campaigns and advice on feeding young children	0.14 (0.24)	0.17 (0.22)	0.17 (0.23)	0.3	0.16 (0.23)
Promotional campaigns and advice on nutrition for women	0.16 (0.33)	0.17 (0.29)	0.16 (0.29)	0.7	0.17 (0.30)

[1] Average (SD)

[2] Kruskal-Wallis tes

Table 6. Exposure to awareness-raising activities by wealth quintile

Awareness-raising activities	Q1 (163) *poor*	Q2 (170)	Q3 (173)	Q4 (170)	Q5 (180) *Rich*	p-valeur[2]
Participation in MERIEM nutrition awareness activities	0.08 (0.28)	0.09 (0.32)	0.12 (0.32)	0.15 (0.37)	0.20 (0.43)	0.002
Fati MARIKO's song on nutrition for pregnant women and young children	0.29 (0.45)	0.44 (0.50)	0.45 (0.50)	0.45 (0.50)	0.48 (0.50)	0.004
Promotional campaigns and advice on feeding young children	0.10 (0.18)	0.14 (0.20)	0.16 (0.21)	0.18 (0.23)	0.22 (0.28)	<0.001
Promotional campaigns and advice on nutrition for women	0.06 (0.09)	0.06 (0.09)	0.06 (0.09)	0.08 (0.10)	0.08 (0.10)	0.2

[1] Average (SD)

[2] Kruskal-Wallis test

The activity used to inform more women was the song by the artist Fati MARIKO on nutrition for pregnant women and young children 0.42 ± 0.49 out of 1. Promotional campaigns with companies producing MERIEM fortified products, followed by advice on diet and nutrition, reached approximately the same number of women. Respectively 0.16 ± 0.23 for the campaigns for the children's product Vitamil+ and 0.17 ± 0.30 for the women's product (Foura soga).

The analysis in Table 6 shows a significant effect of households' socioeconomic status on participation scores in MERIEM awareness-raising activities (Table 6).

Table 7. Distribution of exposure classes for awareness-raising activities according to women's physiological status

Exhibition classes	PW (97)	BW (448)	NPNBW (311)	Total (856)[1]
Not sensitized	41 % (40/97)	35 % (157/448)	34 % (107/311)	36 % (304/856)
Low sensitivity	51 % (49/97)	54 % (242/448)	57 % (176/311)	55 % (467/856)
Moderately aware	6,2 % (6/97)	10 % (46/448)	8,4 % (26/311)	9,1 % (78/856)
Highly sensitized	2,1 % (2/97)	0,7 % (3/448)	0,6 % (2/311)	0,8 % (7/856)

[1] % (n/N)

At the level of all the project's nutrition communication activities, the richer the household, the higher the participation scores of the women in the household. Similarly, the poorer the household, the lower the exposure of the women in the household to the nutrition communication activity. There are highly significant differences (P<0.05) between wealth quintiles according to the non-parametric Kruskal-wallis ANOVA test (Table 6).

Consumers of the fortified product MERIEM Foura Soga are significantly (p<0.001) more exposed to nutrition awareness activities than non-consumers.

However, in both groups, women's exposure to MERIEM project communications was low.

Over 41% of pregnant women surveyed in the city of Niamey are not aware by the MERIEM project, with regard to their own diet and that of their child. More than half of all physiological statuses considered had low levels of awareness, respectively 51% of pregnant woman PW, 54% of breastfeeding woman (BW) and 57% of not pregnant or breastfeeding (NPNB) women (Table 7).

Exposure to MERIEM awareness activities increases from quintile Q1 (poor) to quintile Q5 (affluent). The overwhelming majority of women of childbearing age highly sensitized by the project are in the better-off quintile (Table7).

4| Discussion

4.1|Knowledge of good breastfeeding practices

Breastfeeding is a crucial part of a child's life and is considered one of the most effective strategies for improving health and reducing the risk of morbidity and mortality (Zinho and Affo, 2024). A relatively average level of knowledge of good breastfeeding practices was reported among all women surveyed. The same finding was made by a study in Ghana, which showed a link between maternal knowledge and children's nutritional status (Forh et al., 2022). This result highlights the need to step up nutritional awareness-raising activities among women, emphasizing on the importance and good practices of breastfeeding, in order to improve their knowledge and thus contribute to better child health.

The first food a newborn should ingest is breast milk/colostrum. This answer was given by 76% of consumers and 72% of non-consumers. This result showed that mothers have a good knowledge of the importance of colostrum for their newborns. On the other hand, other women think of zam-zam water, juice or honey as the first food to give to their children. Over 80% of women are familiar with colostrum, and know that it is essential to give it to their children immediately after birth. A study in Nigeria reported that 38% of Nigerian women have a good knowledge of colostrum. Timely initiation of breastfeeding is crucial to positive health outcomes for babies and mothers (Sarfo et al., 2024). Over 70% of women interviewed know that it is recommended to put children to the breast within an hour of birth. A study in Ghana showed significant associations between the child's nutritional status and the time of initiation of breastfeeding (Appoh & Krekling, 2005).

4.2|Knowledge of complementary feeding for young children

The complementary feeding period is a critical period for child growth (Kouton et al., 2017). Women's knowledge of complementary feeding practices was assessed in this study. There is no significant difference between mean knowledge scores of good complementary feeding practices according to consumption status. Delaba et al. (2017) found a significant link between children's nutritional status and mothers' good knowledge of complementary feeding (Debela et al., 2017). Some of the respondents (44%) claim that children can receive semi-solid foods (porridges, purees, soft foods) in addition to breast milk from 6 months. Eighty percent (80%) of women think that children can eat the family dish with other family members before the age of one. The surveyed women's knowledge of complementary feeding was low, with an average score of 2.43 ± 1.45 out of a maximum of 8 points. Yet, the complementary feeding period is a critical period for child growth (Kouton et al., 2017). This is why women's awareness needs to be stepped up at this precise period in their children's development. The main foods cited by women as good for their children over 6 months of age are: milk and dairy products (73%), eggs (54%), legumes and simple cereal flour porridge (33%). As for foods that are not recommended for young children, mothers were most likely to cite tea (39%), rice (26%) and soft drinks (21%).

4.3|Knowledge of children's nutritional status

Women's knowledge of children's nutritional status is relatively good, with an average score of 5.63 ± 1.75 out of a maximum of 7 points. These results are consistent with those reported by Salissou et al., who stated that more than 72% of women have a better understanding of the clinical signs of malnutrition (Salissou et al., 2019). For the majority of WACs surveyed, malnutrition is defined as weight loss or thinning of children (75%). More than one-third of women also think of a lack of energy/weakness: the child is weak or sad and has no strength. More than 80% of women are aware of the birth weight limit of 2.5 kg below which a child is considered to have low birth weight. This finding by the study is very encouraging, although knowledge does not always have a direct effect on the attitudes and practices of this population. A study conducted by Mahamadou et al in the Maradi region revealed that in Niger, the majority of mothers are aware of prematurity but are unaware of the link between the nutritional status of the mother and that of the child (Mahamadou et al, 2022).

4.4|Knowledge of good health and nutrition practices among pregnant women

Knowledge of good health and nutrition practices among pregnant women in Niamey is average, with a mean score of 2.04 ± 0.75 SD out of 4 points. This result highlights nutritional health issues for women and fetuses during pregnancy. More than 70% of these women say that during pregnancy, women should change their eating habits. Mahamadou et al. (2022) reported in their study that 60% of women changed their diet during pregnancy. The main changes cited by women are: eating a varied diet and eating different types of food every day (43%), eating frequently/eating at least meals a day (36%), and eating one more meal a day (35%). The foods most often cited as being good for pregnant women are: fruits and vegetables (53%), legumes (34%), meat, offal, and liver skewers (32%). On the other hand, the foods and beverages most often cited as being inadvisable for pregnant women are: alcoholic beverages (41%), coffee (34%), overly salty foods, salty snacks (31%), and sodas (25%). The foods cited by women largely correspond to the recommendations of Niger's National Nutrition Security Policy (PNSN) (HC I3N, 2021).

4.5|Breastfeeding women's knowledge of good health and nutrition practices

The knowledge of breastfeeding women surveyed in the city of Niamey on good health and nutrition practices is below average, with a score of 1.75 ± 0.77 SD out of 4 points. This result is lower than that reported by Iknane et al. (2011) in Mali, who stated that mothers' level of knowledge about good health and nutrition practices remains acceptable. This low knowledge score could explain these women's poor health and nutrition practices. Women who consumed Foura Soga had significantly better knowledge scores according to the Wilcoxon test. This higher score may be due to their participation in awareness-raising activities.

More than 60% of breastfeeding women say that women need to change their eating habits when breastfeeding. The main changes cited are: eating more (55%), eating frequently (46%), and eating one more meal per day (26%). The foods and beverages most often cited as good for breastfeeding women are: porridge (52%), fish (37%), meat, offal, and liver skewers (34%). On the other hand, the foods and drinks most commonly cited as not recommended for breastfeeding women are: alcoholic beverages (35%), coffee (33%), overly salty foods and salty snacks (31%), and sugary drinks (24%). Most of the foods cited by women are in line with the recommendations of the PNSN in Niger (HC I3N, 2021).

4.6 | Women of childbearing age's knowledge of vitamins, minerals and deficiencies

The best-known vitamin is vitamin A (49%), and it is significantly better known among WCAs who consume the fortified product Foura Soga (69%) than among non-consumers (47%). This result contradicts the findings of Sombie et al. (2023) in Burkina Faso, who stated that vitamin A is very little known among Burkinabe women. B vitamins are known by 22% of the women surveyed. The least known vitamin is vitamin K (2.7%). The best-known minerals are iron (58%) and iodine (12%). More than 45% of women said they knew nothing about vitamins and minerals. A study on the knowledge of women of childbearing age about folic acid showed that women knew little about vitamins and that among those who did know, the vast majority were unaware of their benefits or the recommended period of intake (Salgues et al. 2017).

Based on average scores for knowledge of micronutrients and their deficiencies in the body. The scores are low overall, but the analysis shows that women who consume more than women who don't consume have a better knowledge of micronutrients. According to the Wilcoxon test, the difference was statistically significant for pregnant women and not significant for the other two groups.

4.7 | Surveyed women's participation in nutrition awareness-raising activities

Participation scores for awareness-raising activities reveal low participation by women in awareness-raising activities on nutrition and food issues for women and young children (No more than one in 8 women). Women who use Foura Soga have significantly higher participation scores than non-users (except for pregnant women, for whom the difference is not significant). Breastfeeding women seem to participate more in these activities than pregnant women and women who are neither pregnant nor breastfeeding. The awareness-raising sessions at the health center (with the picture boxes) were the awareness-raising activities in which women had participated the most. In fact, 58% of women had participated in this type of activity at least once. Consumers participated significantly more in this type of activity than non-consumers. No more than 12.7% of women took part in awareness-raising sessions with NGOs. Over 42% of the women we met had heard artist Fati MARIKO's song on nutrition and diet for pregnant women and young children. Also, 40% of the women had already seen or heard promotional campaigns and advice on young children's nutrition.

The channel most used to inform these women was television, which informed 65% of the women surveyed.

4.8 | Women's knowledge according to exposure to awareness-raising activities

The results of this study showed that high exposure to awareness-raising activities significantly improves the nutritional knowledge scores of women of childbearing age. In other words, the more women are exposed to nutritional awareness-raising and education activities, the better their knowledge and attitudes toward nutrition, diet, and child health issues. In a study in France, Desport et al. (2015) also demonstrated that public awareness campaigns on the risks of malnutrition and frailty in older people have a beneficial effect. Knowledge scores are significantly better among highly aware women (P<0.001). This result is consistent with several other studies that have shown a significant effect between the degree of awareness among mothers and improved nutritional practices (Prasetyo et al. 2023). A study in Nigeria clearly demonstrated that nutritional education for mothers has a complementary effect on their knowledge, producing positive results in terms of infant nutrition (Fadare et al. 2019). Studies have examined the effect of nutrition education on mothers' knowledge and children's nutritional status and have reported that nutrition education has had an impact on mothers' knowledge, attitudes, and skills regarding nutrition (Prasetyo et al. 2023).

5 | Conclusion

Overall, this study shows that the WHO guidelines on nutrition for the first 1,000 days of life, concerning the diet of pregnant women, breastfeeding and complementary feeding, are known and more often respected by a large proportion of women, but improvements are still needed to raise awareness among women who are still poorly informed.

Women's participation in nutrition awareness sessions is low. Among awareness-raising activities, the song by the artist Fati Mariko reached the largest number of women (42%) in the survey sample. Many indicators were more favorable among women who consumed Foura Soga (MERIEM fortified product), such as knowledge scores on vitamins, minerals and deficiencies, and those on fortified products, as well as participation scores in awareness-raising activities. Exposure awareness-raising activities improved women's knowledge scores

These findings demonstrate the critical role of maternal knowledge in child health outcomes, and underscore the urgent need for targeted interventions and comprehensive maternal awareness programs to address identified knowledge gaps and improve child health outcomes. The study highlights the positive impact of awareness-raising activities on the nutritional knowledge of WCAs. Looking ahead, it would be important to examine the link between maternal nutritional knowledge and child nutritional outcomes.

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| Conflict of interest

The authors declare that they have no conflict of interest.

| Data availability statement

The data used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

| Ethical approval

This study did not involve any testing or experimentation on humans or animals. No clinical trials or interventions were conducted. The data collected came exclusively from direct questioning of participants, in accordance with local customs and practices and in line with the ethical principles of social research.

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